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Machine Learning Lab

Aim 2_Extra: To implement linear Regression for predicting continous target variables and evaluate the model performance using apporopiate metrics.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
path = '/content/drive/MyDrive/ML_Classes/insurance[1].csv'
df = pd.read_csv(path)
df.head()
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

Next steps: Generate code with df New interactive sheet

```
df.describe()
```

	age	bmi	children	charges
count	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	30.663397	1.094918	13270.422265
std	14.049960	6.098187	1.205493	12110.011237
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.296250	0.000000	4740.287150
50%	39.000000	30.400000	1.000000	9382.033000
75%	51.000000	34.693750	2.000000	16639.912515
max	64.000000	53.130000	5.000000	63770.428010

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   sex       1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

```
df.shape
```

```
(1338, 7)
```

```
from re import X
#step 2 : Feature & Target split

X = df.drop('charges', axis=1) # Independent
y = df['charges']
```

```
X
```

	age	sex	bmi	children	smoker	region
0	19	female	27.900	0	yes	southwest
1	18	male	33.770	1	no	southeast
2	28	male	33.000	3	no	southeast
3	33	male	22.705	0	no	northwest
4	32	male	28.880	0	no	northwest
...	...	...	...	...	...	...
1333	50	male	30.970	3	no	northwest
1334	18	female	31.920	0	no	northeast
1335	18	female	36.850	0	no	southeast
1336	21	female	25.800	0	no	southwest
1337	61	female	29.070	0	yes	northwest

1338 rows × 6 columns

Next steps: Generate code with X New interactive sheet

```
y
```

	charges
0	16884.92400
1	1725.55230
2	4449.46200
3	21984.47061
4	3866.85520
...	...
1333	10600.54830
1334	2205.98080
1335	1629.83350
1336	2007.94500
1337	29141.36030

1338 rows × 1 columns

dtype: float64

```
#Step 3: Train_Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
X_train.shape
```

```
(1070, 6)
```

```
X_test.shape
```

```
(268, 6)
```

```
y_train.shape
```

```
(1070,)
```

```
y_test.shape
```

```
(268,)
```

```
#Step 4: Train Linear Regression Model

# Identify categorical columns
categorical_cols = X_train.select_dtypes(include='object').columns

# Apply one-hot encoding to X_train and X_test
X_train_encoded = pd.get_dummies(X_train, columns=categorical_cols, drop_first=True)
X_test_encoded = pd.get_dummies(X_test, columns=categorical_cols, drop_first=True)

model = LinearRegression()
model.fit(X_train_encoded, y_train)
```

```
▾ LinearRegression
LinearRegression()
```

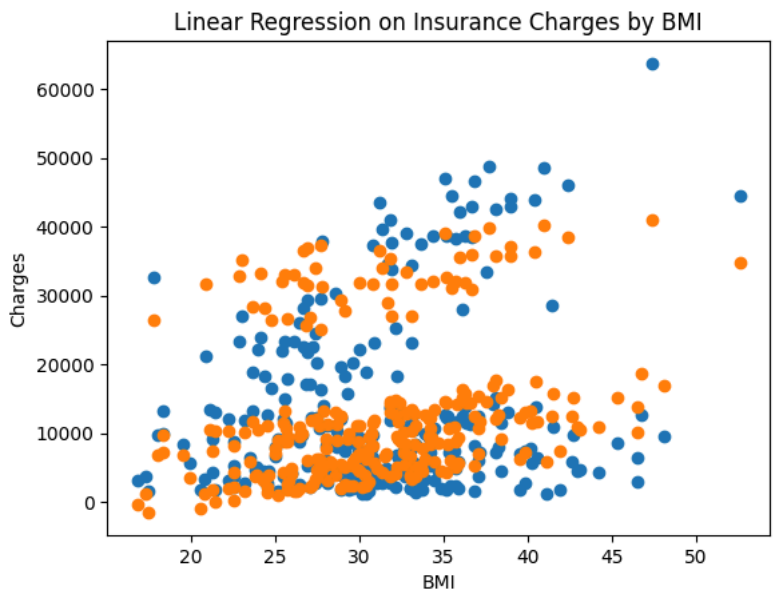
```
#Step 5: Prediction

y_pred = model.predict(X_test_encoded)
```

```
#Step 6: Model Evalution

nae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)
```

```
print("\nModel Evaluation Metrics:")
print("---------------------------")
print("Mean Absolute Error (MAE):", nae)
print("Mean Squared Error (MSE):", mse)
print("Root Mean Squared Error (RMSE):", rmse)
print("R^2 Score:", r2)
```

```
Model Evaluation Metrics:
---------------------------
Mean Absolute Error (MAE): 4181.19447375365
Mean Squared Error (MSE): 33596915.85136145
Root Mean Squared Error (RMSE): 5796.284659276273
R^2 Score: 0.7835929767120724
```

```
# Step 7: Visualization (One Feature)
plt.scatter(X_test['bmi'], y_test)
plt.scatter(X_test['bmi'], y_pred)
plt.xlabel('BMI')
plt.ylabel('Charges')
plt.title('Linear Regression on Insurance Charges by BMI')
plt.show()
```